Stat 266 – Unit 6 – Video Set 4 – Class Examples

Summary:

Theorem 6.1: (Uniqueness)

If the two random variables, X and Y, have the same moment-generating functions, $m_Y(t) = m_X(t)$ for all t, then X and Y have identical probability distributions.

Method of Moment - Generating Functions

Consider $U = h(Y_1, Y_2, \ldots, Y_n)$, a function of the random variables $Y_1, Y_2, \ldots, Y_n$.

- Find the moment-generating function for U, $m_U(t)$.
- Compare $m_U(t)$ with other known moment-generating functions. If $m_U(t) = m_V(t)$ for all values of t, then U and V have identical distributions.

Theorem 6.2: Let $Y_1, Y_2, \ldots, Y_n$ be independent random variables with moment-generating functions $m_{Y_1}(t), m_{Y_2}(t), \ldots, m_{Y_n}(t)$, respectively.

If $U = Y_1 + Y_2 + \cdots + Y_n$ then, $m_U(t) = m_{Y_1}(t) \times m_{Y_2}(t) \times \cdots \times m_{Y_n}(t) = \prod_{i=1}^{n} m_{Y_i}(t)$.

Special Case: If $Y_1, Y_2, \ldots, Y_n$ are also identically distributed with moment-generating function $m_Y(t)$ then,

$$m_U(t) = \left[m_Y(t)\right]^n.$$

Moment Generating functions:

If X is a Random Variables:

Then The moment generating functions is defined to be $m_X(t) = E(e^{tX})$.

Thm 6.1 tell us there is one to one correspondance between X and $m_X(t)$.

How to apply? If X and Y has the same moment Generating functions,

they have the same distributions.

Some example moment generating function:

The Normal distribution: $X \sim N(\mu, \sigma^2)$.

$$E(e^{0+X}) = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{tx} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx. = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2-2\mu x+\mu^2-2t\sigma^2 x}{2\sigma^2}} dx$$

$$= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2 - 2(u+t\sigma^2)x + u^2}{2\sigma^2}} dx = \int_{-\infty}^{+\infty} e^{-\frac{(x-(u+t\sigma^2))^2 + u^2 - (u+t\sigma^2)^2}{2\sigma^2}} dx$$

$$= e^{\frac{(u+t\sigma^2)^2 - u^2}{2\sigma^2}} \underline{\int_{-\infty}^{+\infty} \boxed{\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-(u+t\sigma^2))^2}{2\sigma^2}}} dx}$$

$$\| $$

$$e^{\frac{u^2 + 2ut\sigma^2 + t^2\sigma^4 - u^2}{2\sigma^2}}$$

this is this distribution function of $N(u+t\sigma^2, \sigma)$. Integral it you get 1.

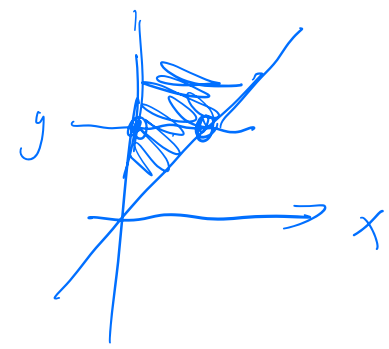
$$= e^{ut + \frac{t^2\sigma^2}{2}}$$

Example A:

Consider two random variables with joint density given as

$$f_{X,Y}(x,y)=\begin{cases}\dfrac{1}{\beta^2}e^{-y/\beta} & , \quad 0<x<y,\\ 0 \ , \quad else\end{cases}$$

Let $U=\dfrac{X}{Y}$. Find $m_U(t)$. What can you conclude about the distribution of U?



$$m_U(t)=E(e^{Ut})=\iint_{0<x<y} e^{\frac{x}{y}t}\cdot\frac{1}{\beta^2}e^{-\frac{y}{\beta}}\,dx\,dy.$$

$$=\frac{1}{\beta^2}\int_0^{+\infty}\int_0^{y} e^{\frac{t}{y}x-\frac{y}{\beta}}\,dx\,dy. = \frac{1}{\beta^2}\int_0^{+\infty}\frac{y}{t}e^{\frac{t}{y}x-\frac{y}{\beta}}\Big|_0^{y}\,dy$$

$$=\frac{1}{\beta^2}\int_0^{+\infty}\frac{y}{t}e^{t-\frac{y}{\beta}}-\frac{y}{t}e^{-\frac{y}{\beta}}\,dy$$

$$=\frac{1}{\beta^2}\left(\frac{e^t}{t}\int_0^{+\infty}ye^{-\frac{y}{\beta}}\,dy-\frac{1}{t}\int_0^{+\infty}ye^{-\frac{y}{\beta}}\,dy\right.$$

$$= \frac{1}{\beta^2}\left(\frac{e^t}{t}\cdot\beta^2 - \frac{\beta^2}{t}\right) = \frac{e^t-1}{t}.$$ This MGF of Uniform distribution.

Example B:

Consider a random variable with moment-generating function $m_Y(t) = e^{\beta t}(1-\beta t)^{-1}$.

Let $U = \frac{Y}{\beta} - 1$. Find $m_U(t)$.

What can you conclude about the distribution of U?

$$m_{\cancel{U}\,Y}(t) = \int e^{ty} f_U(y)\,dy. = e^{\beta t}(1-\beta t)^{-1}$$

$$m_U(t) = \int e^{t(\frac{y}{\beta}-1)} f_U(y)\,dy = \int e^{t\frac{y}{\beta}}\cdot e^{-t} f_Y(y)\,dy$$

$$= e^{-t}\int e^{\frac{y}{\beta}\cdot t} f_Y(y)\,dy = e^{-t}\int e^{\frac{t}{\beta}\cdot y} f_Y(y)\,dy.$$

$$= e^{-t}\cdot m_Y\left(\frac{t}{\beta}\right) = e^{-t}\cdot e^{t}(1-t)^{-1} = (1-t)^{-1}$$

This is the moment Generating function of Exp(1).

Example C:

Let $Y_1,\ldots,Y_n$ be independent and identical *Normal* random variables with $E[Y_i]=\mu$ and $V[Y_i]=\sigma^2$, $i=1,\ldots,n$.

Define $\bar{Y}=\frac{1}{n}\sum_{i=1}^{n}Y_i$.

a) Find the distribution of $\bar{Y}$. What are $E\left[\bar{Y}\right]$ and $V\left[\bar{Y}\right]$?

b) Find the distribution of $Z_i=\frac{Y_i-\mu}{\sigma}$.

c) Find the distributions of Z_i^2.

d) Find the distribution of $X=\sum_{i-1}^{n}Z_i^2$.

a). since $Y_1,\ldots,Y_n$ are independent, Let $U=Y_1+\cdots+Y_n$.

$E_U(t)=m_{Y_1}(t)\cdot m_{Y_2}(t)\cdots m_{Y_n}(t)$ since $Y_i\sim N(\mu,\sigma^2)$.

$m_{Y_i}(t)=e^{\mu t+\frac{t^2\sigma^2}{2}}$

$E_U(t)=\left(e^{\mu t+\frac{t^2\sigma^2}{2}}\right)^n=e^{\mu nt+\frac{t^2 n\sigma^2}{2}}$

Since $\bar{Y}=\frac{U}{n}$ $\quad E_{\bar{Y}}(t)=\int e^{\frac{u}{n}t}f_U(u)\,du.$

$$= \int e^{u\frac{t}{n}} f_u(u)\,du = E_u\left(\frac{t}{n}\right) = e^{ut + \frac{t^2\sigma^2}{2n}}.$$

So $\hat{Y} \sim N\left(u, \left(\frac{\sigma}{\sqrt{n}}\right)^2\right)$.

since $z_i = \frac{y_i - u}{\sigma}$

b). $m_{Y_i}(t) = e^{ut + \frac{t^2\sigma^2}{2}}$ $\qquad m_{z_i}(t) = \int e^{\frac{y_i - u}{\sigma}t} f_{Y_i}(y_i)\,dy_i$

$$= e^{-\frac{u}{\sigma}t} \cdot \int e^{\frac{y_i}{\sigma}t} f_{Y_i}(y_i)\,dy_i$$

$$= e^{-\frac{u}{\sigma}t} \cdot \left(e^{u\frac{t}{\sigma} + \frac{t^2\sigma^2}{2\sigma^2}}\right)$$

$$= e^{\frac{t^2}{2}}$$ this is GMF of $N(0,1)$.

c). since $z_i \sim N(0,1)$ $\quad f_z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$

$$m_{z_i^2}(t) = \int_{-\infty}^{+\infty} e^{z^2 t} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-(\frac{1}{2}-t)z^2} dz.$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{z^2}{2(\frac{1}{1-2t})}} dz$$

$$= \frac{1}{\sqrt{1-2t}} \int_{-\infty}^{+\infty} \underbrace{\boxed{\frac{1}{\sqrt{2\pi \cdot \frac{1}{1-2t}}} e^{-\frac{z^2}{2(\frac{1}{1-2t})}}}}_{\text{Normal distribution}} dz$$

$$= \frac{1}{\sqrt{1-2t}}$$

This is MGF of chi-square distribution of freedom 1.

d). Since Z_i^2 are independent

then $m_X(t) = (m_{Z^2}(t))^n = \left(\frac{1}{\sqrt{1-2t}}\right)^n$

chi-square distribution of freedom n.